

Fraction of Inspired Oxygen (FiO₂)

FiO₂ is the percentage of oxygen available in the air (for breathing). It is a measure of how much oxygen is being delivered to a patient, whether it be natural or through supplemental oxygen devices. The acceptable clinical range for FiO₂ is 21-100 percent. 21% represents atmospheric air (used for breathing at rest). 100% represents pure oxygen (delivered through a ventilator). Any value less than 21% is physiologically impossible for a living persons and a value greater than 100% exceeds the limits of what can be delivered.

A patient breathing room air at 21% is considered stable from a respiratory standpoint. A patient requiring supplemental oxygen between 22-60 percent has moderate respiratory compromise and requires attention. A patient on high flow oxygen/a ventilator at 61-100 percent is critically ill and requires immediate intervention. For this reason, FiO₂ is one of the first vitals assessed during triage. It lets known the severity of a patient's respiratory status to the clinical team.